Alert Triage Pipeline — Validation Report

Controlled simulation validating the automated enrichment & triage pipeline

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1. Summary

An automated alert triage and enrichment pipeline was validated against a controlled six-stage attack simulation on a Windows endpoint. The pipeline correctly parsed, enriched, scored and reported every stage, producing the full range of verdicts (Informational, Suspicious, Malicious) across Sysmon and Windows Security telemetry plus a live Windows Defender detection. All objectives were met.

2. Objective

To confirm that the pipeline, when a qualifying Wazuh alert fires, automatically (a) extracts the alert's indicators, (b) enriches them with threat intelligence, (c) assigns a correct verdict and priority, and (d) delivers an enriched card to the analyst channel.

3. Environment

Wazuh SIEM (192.168.64.6), Windows 11 endpoint DESKTOP-4VD18QN (192.168.64.4), Python pipeline with AbuseIPDB and VirusTotal, Discord delivery. Isolated home lab; no production systems involved.

4. Test Scope

Five telemetry event types plus AV detection were selected to cover the core attacker behaviours. Process and network events were enriched on-demand (high volume); the rest were auto-triaged. This scope was a deliberate design choice to validate correctness across telemetry types without generating noise.

5. Results

#	Stage / event	Rule (level)	Verdict	Enrichment	Result
1	Sysmon 11 - file dropped	92203 (6)	Informational	-	PASS
2	Sysmon 1 - process run	100103 (5)	Informational	VirusTotal (clean)	PASS
3	Sysmon 3 - network connection	100104 (3)	Informational	AbuseIPDB (real IP)	PASS
4	Windows 4720 - account created	60109 (8)	Suspicious	-	PASS
5	Windows 4732 - group changed	60154 (12)	Suspicious	-	PASS
6	Windows Defender - malware detected	62123 (12)	Malicious	Defender verdict	PASS

6. Evidence

The enriched cards produced in Discord for each stage:

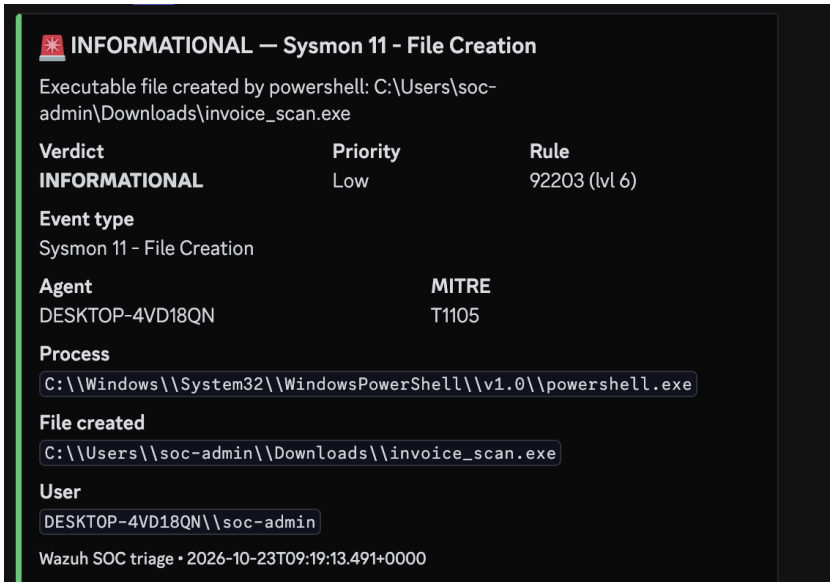


Figure: Stage 1 - Sysmon 11 file creation

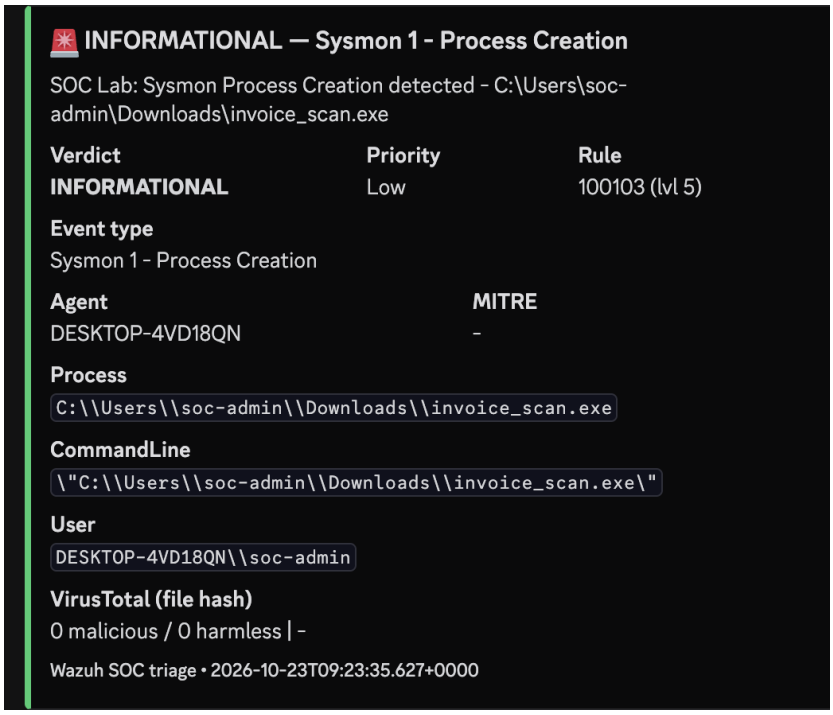


Figure: Stage 2 - Sysmon 1 process + VirusTotal

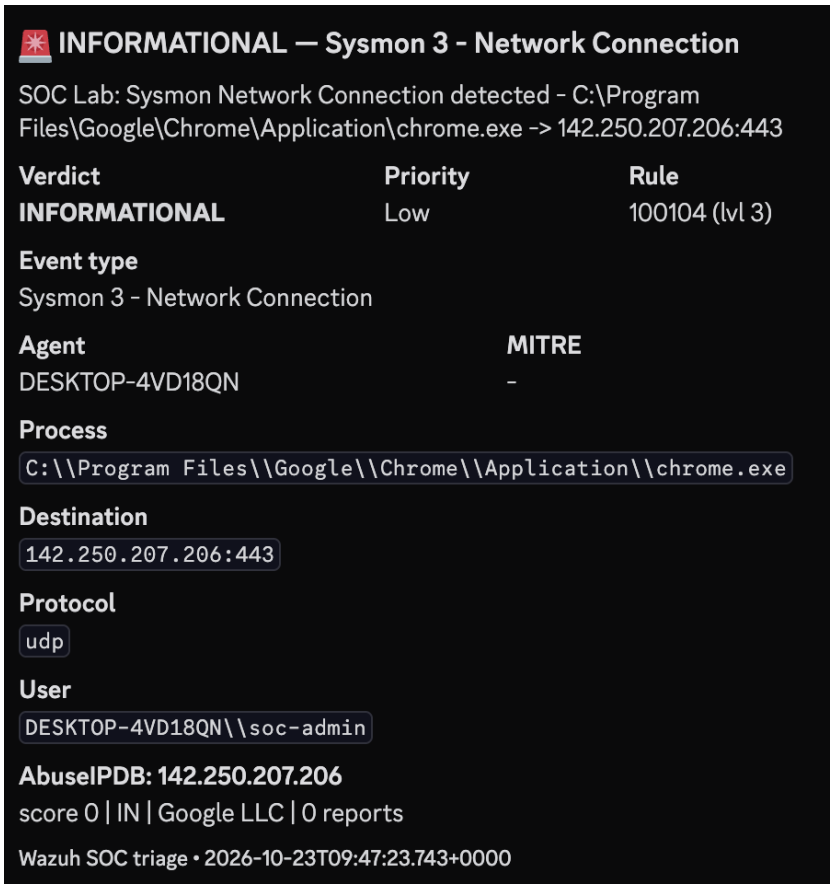


Figure: Stage 3 - Sysmon 3 network + AbuseIPDB



Figure: Stage 4 - Windows 4720 account created

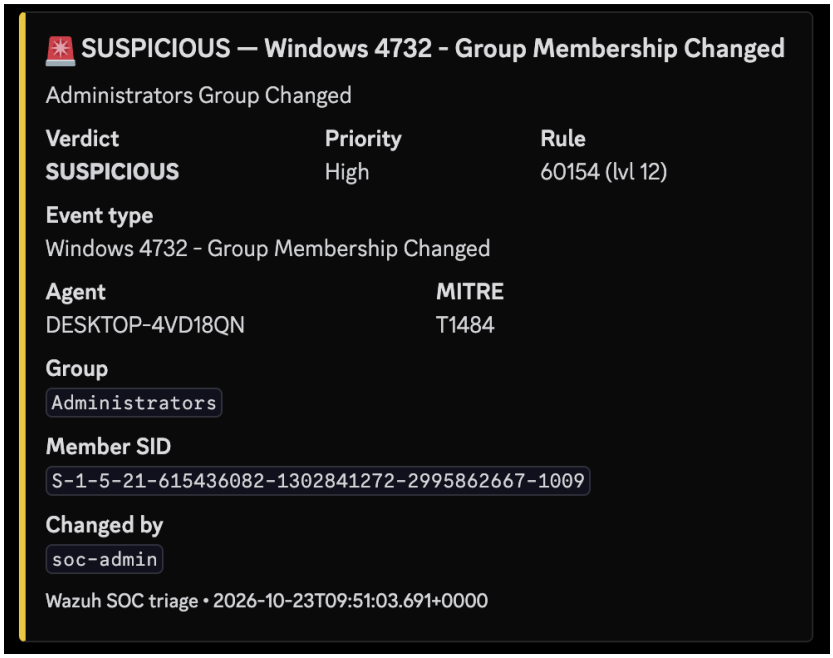


Figure: Stage 5 - Windows 4732 privilege escalation



Figure: Stage 6 - Windows Defender malware detection (MALICIOUS)

7. Observations & Findings

- Automatic triggering worked reliably server-side (unlike the agent-side Active Response in Project 1).
- Threshold tuning: the file-drop fired rule 92203 at level 6, below the initial level-8 trigger; the threshold was lowered to level 6 to include file-creation events.
- Endpoint noise: process and network events fire constantly; the on-demand model for these is the correct approach and avoids channel flooding and API-quota exhaustion.
- Enrichment returned real data: AbuseIPDB gave country/ISP/score for a public IP; VirusTotal returned 60+ detections on the EICAR hash during validation of the malicious path.

8. Limitations & Next Steps

- Validated with known indicators (EICAR hash, real public IP, live Defender detection) because lab indicators are not in public feeds.
- Domain/URL enrichment (via DNS Event 22) and additional threat-intel sources are planned next steps.
- Automated response actions (disable account / block IP on malicious verdict) are a natural extension, building on Project 1's Active Response.

9. Conclusion

The pipeline met every objective: it automatically enriched and triaged alerts across six telemetry types, produced correct verdicts and priorities, and delivered analyst-ready cards in real time. Note: the Python

implementation was written with AI assistance; the design, integration, testing and validation were carried out by the author.